

Daily Logic Drill #6

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Mathematical Induction · Base Case Traps · Inductive Step Mechanics · Misapplied Induction.

Attempt each question before checking answers. Time yourself. No calculators.

Section A Rapid Recognition

(≤15 s each)

Induction structure, base cases, and inductive steps. Pure recognition.

1. What are the two essential components required for a proof by mathematical induction?
2. To prove a statement $P(n)$ holds for all integers $n \geq 4$ by induction, what is the base case?
3. If $P(k) \implies P(k+1)$ is proved, but no base case $P(n_0)$ is established, is $P(n)$ proved for any n ?
4. What is the inductive hypothesis in a proof of $\sum_{i=1}^n i = \frac{n(n+1)}{2}$?
5. True or false: “Mathematical induction can be used to prove statements about all real numbers x .”
6. In a proof that $2^n > n^2$ for $n \geq 5$, what is the smallest integer n for which the base case must be checked?
7. What is the negation of the inductive step statement: “For all $k \geq 1$, $P(k) \implies P(k+1)$ ”?
8. True or false: “If $P(1)$ is true and $P(k) \implies P(k+2)$ for all $k \geq 1$, then $P(n)$ is true for all positive integers n .”
9. For which positive integers n is $2^n > n$ true?
10. What is the base case needed to prove $n! > 2^n$ by induction?

Section B Manipulation Drills

(≤50 s each)

Algebraic manipulations within inductive steps.

1. Express $\sum_{i=1}^{k+1} i^2$ in terms of $\sum_{i=1}^k i^2$.
2. Assuming $\sum_{i=1}^k i = \frac{k(k+1)}{2}$, add $(k+1)$ and simplify to show $\sum_{i=1}^{k+1} i = \frac{(k+1)(k+2)}{2}$.

3. Prove by induction: $\sum_{i=1}^n (2i - 1) = n^2$ for all positive integers n .
4. Assuming $3^{2k} - 1$ is divisible by 8, express $3^{2(k+1)} - 1$ in a form that shows it is also divisible by 8.
5. Prove by induction: $2^n > n$ for all positive integers n .
6. Assuming $k! > 2^k$ for some $k \geq 4$, prove $(k + 1)! > 2^{k+1}$.
7. Prove by induction that $5^n - 1$ is divisible by 4 for all positive integers n .
8. Assuming $\sum_{i=1}^k 2^i = 2^{k+1} - 2$, show that $\sum_{i=1}^{k+1} 2^i = 2^{k+2} - 2$.
9. Prove by induction that the sum of the interior angles of a convex n -gon is $(n - 2) \times 180^\circ$ for $n \geq 3$.
10. Show that if $P(k) \implies P(k + 2)$ is proved, established base cases for both $n = 1$ and $n = 2$ are sufficient to prove $P(n)$ for all $n \geq 1$.

Section C Substitution & Structure

(≤ 90 s each)

Analyze induction fallacies and non-standard inductive setups with tricky MCQs.

1. A student attempts to prove $P(n)$: “ $n^2 + 5n + 1$ is even” for all $n \geq 1$.
 Inductive Step: Assume $P(k)$ is true, so $k^2 + 5k + 1 = 2m$.
 Then $(k + 1)^2 + 5(k + 1) + 1 = k^2 + 2k + 1 + 5k + 5 + 1 = (k^2 + 5k + 1) + 2k + 6 = 2m + 2(k + 3) = 2(m + k + 3)$, which is even. So $P(k) \implies P(k + 1)$ holds!
 However, $P(1) = 1 + 5 + 1 = 7$, which is odd!
 Which evaluation of this situation is correct? (after TMUA 2022 Paper 2 Q14)
 - A) The proof is complete because the inductive step was proven.
 - B) $P(n)$ is false for all $n \geq 1$ despite the inductive step $P(k) \implies P(k + 1)$ being completely valid.
 - C) The inductive step $P(k) \implies P(k + 1)$ contains an algebraic error.
 - D) $P(n)$ is true for all $n \geq 2$.
 - E) Mathematical induction cannot be applied to parity statements.
2. Consider the claim $P(n)$: “ $2^n > n^2$.” (after SMC 2018 Q22)
 Base Case checking: $n = 1 \implies 2 > 1$ (True); $n = 2 \implies 4 > 4$ (False); $n = 3 \implies 8 > 9$ (False); $n = 4 \implies 16 > 16$ (False); $n = 5 \implies 32 > 25$ (True).
 For which set of positive integers n is $2^n > n^2$ actually true?
 - A) $n = 1$ and all $n \geq 5$
 - B) all $n \geq 1$
 - C) all $n \geq 5$ only
 - D) $n = 1, 2, 3, 4, 5$
 - E) no positive integers n

3. A student claims: “To prove $P(n)$ for all $n \geq 1$, it is sufficient to prove $P(1)$ and $P(k) \implies P(2k)$ for all $k \geq 1$.”
Is this claim true or false?
- A) True, because powers of 2 cover all positive integers.
B) False, because this only proves $P(n)$ for n of the form 2^m ($m \geq 0$).
C) True, by the Archimedean property.
D) False, because $P(1)$ cannot be a base case.
E) True, this is known as Cauchy induction.
4. Consider the Fibonacci sequence $F_1 = 1, F_2 = 1, F_3 = 2, F_4 = 3, \dots$ defined by $F_{n+1} = F_n + F_{n-1}$.
What is the value of $\sum_{i=1}^n F_i$?
- A) $F_{n+1} - 1$
B) $F_{n+2} - 1$
C) F_{n+2}
D) $F_{n+1} + 1$
E) $F_{n+3} - 2$
5. A student attempts to prove: “For all positive integers n , $n! \leq 2^n$.”
Checking cases: $n = 1 \implies 1 \leq 2$ (True); $n = 2 \implies 2 \leq 4$ (True); $n = 3 \implies 6 \leq 8$ (True);
 $n = 4 \implies 24 \leq 16$ (False).
Which of the following is true regarding $n!$ vs 2^n ?
- A) $n! \leq 2^n$ for all $n \geq 1$.
B) $n! \leq 2^n$ for $n \in \{1, 2, 3\}$, but $n! > 2^n$ for all $n \geq 4$.
C) $n! > 2^n$ for all $n \geq 1$.
D) $n! \leq 2^n$ for all even n .
E) $n!$ and 2^n are equal for all $n \geq 4$.
6. Consider the sum $S_n = \frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \dots + \frac{1}{n(n+1)}$.
Using mathematical induction, what is the closed-form expression for S_n ?
- A) $\frac{n}{n+1}$
B) $\frac{1}{n+1}$
C) $\frac{n+1}{n}$
D) $\frac{n}{2(n+1)}$
E) $\frac{n^2}{n+1}$
7. To prove $P(n)$ for all integers $n \geq 1$ using Strong Induction, what is the inductive hypothesis step?
- A) Assume $P(k)$ is true for a single integer k .
B) Assume $P(1), P(2), \dots, P(k)$ are all true, and deduce $P(k+1)$.

- C) Assume $P(k+1)$ is true and deduce $P(k)$.
- D) Assume $P(k)$ is false and deduce a contradiction.
- E) Assume $P(n)$ holds for all real numbers $n \leq k$.

8. For which positive integers n is $n^2 < 2^n$ false?

- A) $n = 2, 3, 4$
- B) $n = 1, 5, 6$
- C) $n = 2$ only
- D) $n = 3$ only
- E) $n = 4$ only

Section D Challenge

(SMC / BMO1 difficulty)

Subtle induction proofs and classic competition induction problems.

1. Prove by induction: Bernoulli's Inequality states that for all $x > -1$ and all positive integers n , $(1+x)^n \geq 1+nx$. At which step in the inductive transition $(1+x)^k \geq 1+kx \implies (1+x)^{k+1} \geq 1+(k+1)x$ is the condition $x > -1$ strictly required?
 - A) When establishing the base case $n = 1$.
 - B) When multiplying both sides of $(1+x)^k \geq 1+kx$ by $(1+x)$ without flipping the inequality sign.
 - C) When expanding $(1+kx)(1+x) = 1+(k+1)x+kx^2$.
 - D) When dropping the term $kx^2 \geq 0$.
 - E) The condition $x > -1$ is never required.
2. Prove by induction that $a^n - b^n$ is divisible by $a - b$ for all positive integers n , where a, b are distinct integers. Show the inductive step using the identity $a^{k+1} - b^{k+1} = a(a^k - b^k) + b^k(a - b)$.
3. Consider a grid of $2^n \times 2^n$ squares with one single 1×1 square removed. Prove by mathematical induction that for every integer $n \geq 1$, the remaining region can be completely tiled by L-trominoes (shapes made of 3 squares forming an L).
4. Let $a_1 = 1$ and $a_{n+1} = \sqrt{2 + a_n}$ for $n \geq 1$. (a) Prove by induction that $a_n < 2$ for all $n \geq 1$. (b) Prove by induction that $a_n < a_{n+1}$ for all $n \geq 1$.
5. Prove by induction that $\sum_{i=1}^n i^3 = (\sum_{i=1}^n i)^2 = \left[\frac{n(n+1)}{2}\right]^2$ for all positive integers n .

"Speed comes from recognition, not from rushing. Every shortcut is a pattern you have internalised."

Found a mistake or want to contribute a sheet?

Visit the open-source project at github.com/The-CerealDev/speed-maths